Expressive piano continuation

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Task and implementation

A local MIDI application continues a solo-piano prompt using an LSTM, a scratch Transformer or an unmodified pretrained Aria reference. It exports editable MIDI and fixed-piano audio, preserves prompt notes and reports actual continuation extent, stopping reason and release tail.

Data and training

MAESTRO v3: 1,276 parsed performances, 1,262 retained (957 train, 134 validation, 171 test). Conservative exclusions preserve official splits and prevent known metadata/hash overlap. The pinned 17,727-token representation uses 10 ms timing, five-second segments and duration limits. Baselines share context 2,048, BF16 and 100M targets. LSTM: 9.62M parameters, one 256-unit layer. Transformer: 36.36M parameters, eight 512-wide layers and eight heads.

Scratch checkpoint	Scored targets	Validation loss
LSTM baseline	100,003,399	4.316104
Transformer baseline	100,003,399	3.275785
Transformer continued	150,007,166	3.193548
Transformer musical final	160,021,924	3.178622

Loss is nats on 536 unchanged validation crops and 1,093,516 targets. Baselines share budgets but differ in size. Later runs add compute; musical masking changes the scored-target objective. No test data informed fitting or selection.

Fine-tuning and evidence

The musical run adds 10.01M scored targets using coherent key/time augmentation on half the crops and supplied-prompt continuation loss on three quarters. Raw loss improves 0.47%; conditional next-pitch accuracy rises from 25.31% to 26.00%. All 63 tests and all 16 new comparison generations passed. One listener judged eight earlier pairs: five continued preferences, two original, one unclear; one preference followed reveal. Fresh candidate listening remains pending.

Limits and equal contributions

Predictive gains do not establish audible coherence. Scope is piano, with finite context and quantized timing/durations. No project Aria LoRA result, final test evaluation, broad listening study or human rehearsal is claimed. Latest browser visual/playback inspection remains unverified. Zahoor: data/tokenization, scratch training and checkpoint/Transformer experiments. Ibrahim: Aria integration, audio, application, listening pages and packaging. Both share interpretation and demo preparation. Development used an AI coding assistant.

Private repository: github.com/zahoormzm/expressive-piano-continuation. README supplies setup/run/recovery instructions; the private release supplies trained artifacts. Sources: MAESTRO (Hawthorne et al., ICLR 2019), Music Transformer (Huang et al., ICLR 2019), Aria (Bradshaw et al., ISMIR 2025). Recorded metrics and licenses are in docs and README.